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The Impact of Digital Technology Use on Teaching Quality in University Physical Education: An Interpretable Machine Learning Approach

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Abstract

Amid the ongoing digital transformation of higher education, increasing attention has been paid to the impact of digital technologies on teaching quality—particularly in physical education settings that require high levels of interaction and physical engagement. This study examined 1158 university students across China using a cross-sectional design, integrating interpretable machine learning models with structural equation modeling to systematically assess how the frequency of use of seven common digital technologies influences teaching quality in physical education classes. The study also investigated the mediating roles of perceived teacher support and academic self-efficacy. Nine machine learning models and logistic regression analyses were conducted to predict teaching quality, employing feature permutation importance and SHapley Additive exPlanations (SHAP) to evaluate the relative contribution of each digital tool. Results identified multimedia courseware, e-textbooks, and question banks as the most influential predictors of teaching quality. The SHAP analysis further revealed that management software, educational websites, and multimedia materials had significant positive effects and exhibited synergistic interactions in improving teaching outcomes. Structural equation modeling showed that digital technology use had a significant direct positive effect on teaching quality (B = 0.347, $p < 0.001$). Moreover, perceived teacher support and academic self-efficacy not only served as independent mediators (95% CI = [0.146, 0.226], $p < 0.001$; 95% CI = [0.024, 0.063], $p < 0.001$) but also functioned as a chain mediating effect (95% CI = [0.036, 0.083], $p < 0.001$). This study is the first to integrate interpretable machine learning with structural modeling to elucidate the mechanisms through which digital technologies influence teaching quality in university-level physical education. The findings underscore the critical mediating roles of teacher support and student self-efficacy, providing both theoretical contributions and practical implications for enhancing digital pedagogy in higher education.

Keywords: digital technology; physical education; teaching quality; interpretable machine learning; perceived teacher support; academic self-efficacy

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1. Introduction

In higher education, teaching serves as the foundation and core function of academic institutions. High-quality instruction not only facilitates knowledge exchange and technology transfer but also attracts and cultivates talented students, thereby enhancing institutional reputation and contributing to national innovation capacity. With the rapid advancement

of science and technology, instructional tools and pedagogical methods have continuously evolved, with various forms of digital technology increasingly integrated into classroom instruction [1]. In many countries including the United States and Germany, multimedia education systems have been widely applied in teaching practice [2]. In China, many universities have implemented digital learning platforms and learning management systems to assess student performance, learning preferences, and classroom engagement, thereby enabling more individualized support and attention to diverse learner needs [3]. Additionally, the adoption of virtual classrooms, e-books, and educational websites has improved the accessibility and equity of educational resources, supporting pre-class preparation, in-class participation, and post-class review [4].

In China, undergraduate physical education programs are generally structured as four-year, full-time degree programs governed by national standards set by the Ministry of Education. Nearly all universities and colleges offer such programs. These curricula are designed not only to prepare students to become qualified physical education teachers at the primary, secondary, and tertiary levels, but also to train individuals with advanced expertise to serve as professional coaches for competitive sports teams. Although the national curriculum provides a standardized framework, regional adaptations may occur based on local educational priorities or institutional capacities. The use of multimedia technologies—such as video and audio—has been shown to enhance students' comprehension and retention of instructional content, particularly in subjects with high levels of dynamism and interactivity, such as physical education [5]. Physical education classes often involve visual and experiential components, including movement demonstrations, peer interaction, and skill-based training. Integrating multimedia tools into physical education settings enables students to understand concepts and techniques more intuitively, thereby increasing engagement and improving instructional quality [6,7]. Furthermore, digital technologies such as wearable devices and activity trackers provide real-time data on key physiological indicators (e.g., heart rate, speed, and step count). These tools not only help students monitor their physical condition and prevent injuries but also equip instructors with data to design personalized instructional plans, thereby improving physical education teaching effectiveness [8].

The adoption of interpretable machine learning has significantly improved predictive accuracy and enhanced transparency in educational research, particularly in analyzing student learning processes. Studies employing machine learning models have demonstrated that factors such as the frequency of physical education classes and the degree of digitalization in home and school environments positively influence Chinese secondary students' engagement in digital physical education activities [9]. In the context of physical education, interpretable machine learning models have also been applied to specific sports domains, including basketball training, martial arts techniques, and badminton stroke analysis [10,11]. Despite the expanding use of digital technologies in physical education, the empirical application of interpretable machine learning to instructional research in this area remains in its infancy. In particular, systematic investigations into the differentiated and synergistic effects of various digital tools on teaching quality—through the lens of interpretable machine learning—are still notably lacking.

Perceived teacher support refers to students' subjective perceptions and emotional experiences of the support provided by their instructors, encompassing academic, emotional, and competence-based dimensions [12]. According to self-determination theory, teacher support fulfills students' basic psychological needs for autonomy, competence, and relatedness, thereby fostering intrinsic motivation and classroom engagement, which in turn enhance students' perceptions of teaching quality [13]. In this study, teaching quality is defined as students' perceived quality of the instructional process in smart classroom

environments, including dimensions such as collaboration and communication, inquiry and critical thinking, emotional belonging, and classroom order and discipline. Empirical research indicates that when students perceive strong support and encouragement from teachers, they exhibit greater motivation, higher engagement in learning activities, and improved academic outcomes [14]. Beyond behavioral engagement, perceived teacher support also enhances students' interest and enjoyment in learning, enriching both their cognitive and emotional experiences and ultimately contributing to higher instructional quality [15]. Conversely, a lack of teacher support has been linked to lower academic engagement, reduced confidence, and poorer academic performance [16]. The widespread integration of digital technologies into education has enabled teachers to deliver content and provide encouragement through various channels and formats [17]. Tools such as learning platforms and learning management systems facilitate real-time feedback, promote interactive and flexible instruction, and support individualized learning, thereby strengthening students' perceptions of teacher support [18]. In addition, supportive teacher behaviors have been shown to significantly increase students' acceptance of digital tools, motivation to learn, and classroom engagement—all of which contribute to improved teaching quality [19]. However, recent empirical research on online learning has warned that excessive reliance on digital technologies, combined with limited face-to-face interaction, may diminish students' sense of emotional connection with teachers. Over time, this can lead to feelings of isolation and academic burnout, ultimately impairing learning outcomes [17].

Academic self-efficacy refers to students' beliefs about their academic capabilities and their confidence in successfully completing specific academic tasks [20]. According to self-efficacy theory, students who believe in their ability to succeed are more likely to engage actively in learning and achieve positive academic outcomes. Repeated academic success is considered the most direct pathway to strengthening self-efficacy [21]. In classroom settings, students can access real-time feedback and guidance through digital learning platforms, online resources, educational software, and virtual simulation technologies. These tools allow learners to create personalized study plans aligned with their individual needs, thereby enhancing their autonomy and control over the learning process and, in turn, reinforcing academic self-efficacy [22]. A study in Spain found that higher digital literacy was positively associated with elevated levels of academic self-efficacy among students [23]. Additionally, students who frequently use digital technologies outside the classroom for academic purposes tend to report significantly greater self-efficacy than those who seldom use such tools [24]. Extensive research has shown that students with high academic self-efficacy demonstrate greater effort and persistence in their studies, and that self-efficacy reliably predicts academic performance. In contrast, students with low self-efficacy are more likely to avoid or withdraw from challenging tasks [25]. In the context of physical education, academic self-efficacy plays an especially critical role. Evidence suggests that students with higher self-efficacy are more likely to actively participate in physical activities and derive enjoyment from the learning process [26]. Moreover, in digital and virtual learning environments, academic self-efficacy has been found to increase student engagement and improve both instructional quality and academic outcomes [27].

Social cognitive theory posits that human behavior results from dynamic interactions among environmental, personal, and behavioral factors [28]. In the context of digitalized physical education, teacher support functions as a key environmental factor shaping students' academic beliefs and perceptions, thereby influencing their learning behaviors and instructional experiences. Empirical evidence indicates that when teachers provide constructive feedback, set clear expectations, offer timely guidance, and extend emotional encouragement, they strengthen students' confidence in their academic abilities, thus

enhancing academic self-efficacy [29]. Studies among Spanish university students have shown that perceived teacher support significantly predicts academic self-efficacy [29], a finding echoed in research conducted in Asian educational contexts. In Chinese higher education, academic support from instructors has been found to significantly enhance students' perceived academic competence, especially when facing challenging tasks [30]. Moreover, emotional support—such as encouragement and empathy—has been shown to be particularly beneficial for students with low academic self-efficacy [31]. Research in physical education classrooms suggests that teacher support fosters academic self-efficacy, which in turn promotes positive emotional states and increased classroom engagement—both of which are essential for teaching quality [32]. Evidence from smart classroom environments further indicates that perceived teacher support and academic self-efficacy are key drivers of deep cognitive engagement [33].

The integration of digital technologies into university classrooms has become a global educational trend. These tools enable teachers to provide real-time feedback and personalized instructional support, thereby enhancing students' perceptions of teacher support [34]. In turn, perceived teacher support plays a crucial role in shaping students' academic self-efficacy. Students with higher levels of self-efficacy are more likely to engage actively in learning and classroom interactions, ultimately improving instructional quality [35].

Based on this theoretical and empirical foundation, the following hypotheses are proposed (see Figure 1):

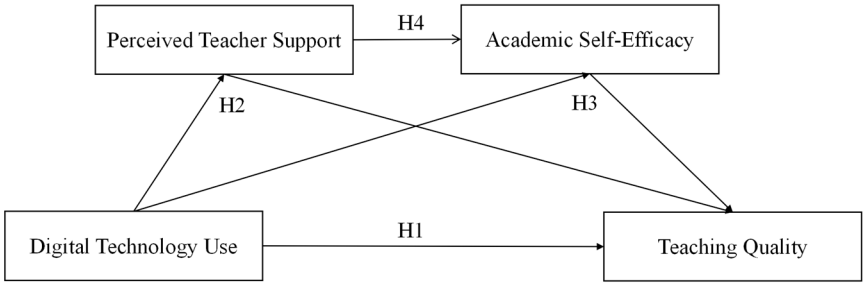

Figure 1. Hypothetical model.

Hypothesis 1: *Digital technology use has a significant positive impact on teaching quality.*

Hypothesis 2: *Perceived teacher support mediates the relationship between digital technology use and teaching quality.*

Hypothesis 3: *Academic self-efficacy mediates the relationship between digital technology use and teaching quality.*

Hypothesis 4: *Perceived teacher support and academic self-efficacy jointly serve as a sequential mediating mechanism in the relationship between digital technology use and teaching quality.*

Given the potentially complex nonlinear relationships and interaction effects among multiple digital technology tools and teaching quality, traditional linear or logistic regression models may be inadequate for capturing the full influence of predictor variables. To address this limitation, the present study employed and compared multiple machine learning models to improve predictive accuracy and identify the most influential variables. Unlike conventional statistical methods, machine learning models offer enhanced capabilities for modeling nonlinear associations, greater robustness, and automated feature selection—advantages particularly suited to high-dimensional data and contexts involving intricate variable interdependencies. To increase model interpretability, the study incorpo-

rated SHapley Additive exPlanations (SHAP) and permutation feature importance analyses to visualize the relative contributions of each digital tool to the prediction of teaching quality. This approach not only yielded high predictive performance but also provided deeper insights into the mechanisms through which digital technologies affect instructional outcomes, offering practical implications for evidence-based educational decision-making.

2. Materials and Methods

2.1. Participants and Procedure

This study employed a cross-sectional design and utilized a combination of convenience and snowball sampling to recruit university students across China. Data were collected from 1–31 December 2024, via the online survey platform Wenjuanxing (https://www.wjx.cn (accessed on 1 December 2024)). Participants completed a structured questionnaire assessing their use of digital technologies in physical education classes during the previous semester, perceived teacher support, academic self-efficacy, and perceived teaching quality. A total of 1612 questionnaires were distributed. After excluding responses completed in under 15 min, 1158 valid responses were retained, resulting in a valid response rate of 71.84%. All participants provided informed consent, acknowledging the voluntary nature of their participation as well as the anonymity and confidentiality of their responses. The study complied with the Declaration of Helsinki and was approved by the Ethics Committee of the School of Basic Medical Sciences, Shandong University (Approval No. ECSBMSSDU2022-1-086).

A total of 1158 participants were included in the study, all of whom were current students enrolled in four-year undergraduate programs and had prior experience using digital technologies in physical education classes. Of these, 717 were male (61.92%) and 441 were female (38.08%). A total of 1043 respondents (90.07%) identified as Han ethnicity, while 115 (9.93%) identified with other ethnic groups. Participants were distributed across undergraduate academic years as follows: 653 were first-year students (56.39%), 402 were in their second year (34.72%), 91 were in their third year (7.86%), and 12 were in their fourth year (1.04%). Regarding geographical distribution, 393 students (33.94%) were from East China, 183 (15.80%) from South China, 329 (28.41%) from Central China, 10 (0.86%) from North China, 1 (0.09%) from Northwest China, 84 (7.25%) from Southwest China, and 158 (13.64%) from Northeast China.

2.2. Measures

2.2.1. Use of Digital Technology

The use of digital technology in physical education classrooms was assessed using an adapted version of the "Actual Behavior" subscale from the Digital Educational Resource Utilization Scale for Teachers [36]. This instrument comprises two dimensions: the use of digital devices and the use of digital resources, and evaluates the frequency of classroom use across nine specific digital tools. A five-point Likert scale was used, ranging from 1 (never) to 5 (very frequently). The scale demonstrated excellent internal consistency, with a Cronbach's alpha coefficient of 0.972.

2.2.2. Perceived Teacher Support

Perceived teacher support was measured using a questionnaire developed by Yangdan, based on earlier research on differential teacher behaviors [37], expert consultation, and classroom observation [38]. The instrument consists of three dimensions: academic support, emotional support, and competence support. It includes 19 items, each rated on a five-point Likert scale from 1 (strongly disagree) to 5 (strongly agree). A total score was calculated by summing all item responses, with higher scores indicating greater perceived teacher

support. The internal consistency of the scale was high in this study, with a Cronbach's alpha of 0.939.

2.2.3. Academic Self-Efficacy

Academic self-efficacy was assessed using a scale developed by Pintrich and DeGroot (1990) [39] and subsequently adapted by Liang [40]. The scale comprises two subscales: self-efficacy for learning ability and self-efficacy for learning behavior. The first subscale measures students' perceived ability and confidence in completing academic tasks, achieving high grades, and avoiding failure. The second evaluates students' confidence in employing effective learning strategies to attain academic goals. Each subscale contains 11 items, for a total of 22 items. Responses were rated on a five-point Likert scale, and a composite score was calculated, with higher scores indicating stronger academic self-efficacy. In the current study, the scale exhibited strong internal consistency (Cronbach's $\alpha = 0.927$).

2.2.4. Teaching Quality

Teaching quality was evaluated using a revised version of Marsh's Teaching Effectiveness Questionnaire [41], which includes three dimensions: student negotiation, inquiry learning, and reflective thinking, totaling 24 items. Responses were collected using a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The scale aimed to assess students' perceptions of instructional quality within smart classroom environments, with an emphasis on student–student and teacher–student interactions, as well as engagement during digital technology integration. The scale showed excellent internal consistency, with a Cronbach's alpha of 0.993.

2.3. Construction and Evaluation of Interpretable Machine Learning Models

Compared to logistic regression, machine learning algorithms generally offer superior capabilities in modeling nonlinear relationships, exhibit stronger predictive performance, possess built-in mechanisms for assessing feature importance, and demonstrate greater fault tolerance and robustness. To ensure methodological comprehensiveness and enhance the robustness of our findings, we selected nine machine learning models representing diverse algorithmic families: linear models (LR, RR), tree-based models (DT, RF, and GBDT), kernel-based models (SVM), instance-based models (KNN), ensemble strategies (AB and VC), and deep learning architectures (MLP). This selection enabled a systematic comparison across models varying in interpretability, capacity for nonlinear modeling, and sensitivity to data structure. For reference, a logistic regression model was also employed as a baseline. All models were implemented using Python 3.10.9. The dataset was partitioned into a training set (80%) and a testing set (20%), comprising 926 and 232 samples respectively. Hyperparameters for each model were optimized via grid search with five-fold cross-validation on the training set. To ensure reproducibility, random seeds were fixed during model initialization. Model performance and robustness were assessed using several evaluation metrics, including the area under the receiver operating characteristic curve (AUC), accuracy, and sensitivity/recall. Hyperparameter configurations for each model are provided in Table S1. To enhance model interpretability, the study employed both permutation feature importance and an SHAP analysis. Permutation importance quantified the contribution of each feature to the model's predictive accuracy. The SHAP analysis offered a comprehensive view by providing both global insights into feature impact and directionality, as well as local explanations of how individual feature values influenced model output. This combined approach also allowed for the identification of threshold effects in feature contributions.

2.4. Statistical Analysis

All statistical analyses were performed using Python 3.10.9, SPSS 26.0, and Amos 28.0 (IBM Corp., Armonk, NY, USA). A Pearson correlation analysis was conducted to examine relationships among all study variables. To identify the most relevant digital technology features and assess multicollinearity, least absolute shrinkage and selection operator (LASSO) regression and variance inflation factor (VIF) analyses were employed. Teaching quality was dichotomized using a median split for use in classification tasks within machine learning models. In contrast, within the structural equation modeling and mediation analyses, teaching quality was retained as a continuous variable. The fit of the structural equation model (SEM) was evaluated using multiple indices: chi-square (χ^2), goodness-of-fit index (GFI), adjusted goodness-of-fit index (AGFI), comparative fit index (CFI), Tucker–Lewis index (TLI), normed fit index (NFI), incremental fit index (IFI), and root mean square error of approximation (RMSEA). Mediating effects were assessed using the PROCESS macro (models 4 and 6) in SPSS, with a bias-corrected bootstrap estimation based on 5000 resamples to compute confidence intervals. A mediating effect was considered statistically significant if the 95% confidence interval did not include zero. For all other statistical tests, significance was set at $p < 0.05$.

3. Results

3.1. Analysis of Variance and Variable Selection

Table S2 displays the results of the analysis of variance (ANOVA) examining the associations between each digital technology tool and teaching quality. Significant differences in teaching quality were found based on the frequency of use of several tools in physical education classes, including projection screens, mobile devices, multimedia courseware, multimedia materials, question banks, subject software, websites, e-textbooks, and management software ($p < 0.001$). LASSO regression analysis—with the optimal regularization parameter determined via cross-validation ($\alpha = 0.00319$)—indicated that both projection screens and mobile devices contributed negligibly to the predictive model, as their coefficients were penalized to zero (Figure S1). Consequently, these two variables were excluded from subsequent model testing. A VIF analysis confirmed that all remaining variables had values below 10, indicating no multicollinearity.

3.2. Model Performance Evaluation

To evaluate classification performance in predicting teaching quality, nine machine learning models and one logistic regression model were systematically compared using receiver operating characteristic (ROC) curves, learning curve, confusion matrices, and performance metrics including accuracy and F1 score (Figures 2, S2 and S3, and Table 1). Among all models, RR achieved the highest performance, with an AUC of 0.75 and an accuracy of 77.6%, outperforming traditional logistic regression (accuracy = 76.3%). Therefore, the RR model was selected for further analysis. As shown in Figure 3, the top three features based on F1 score were multimedia courseware, question banks, and mobile devices/websites (tied).

3.3. Interpretable Modeling Approaches

3.3.1. Ranking the Importance of Digital Technology Use

Feature importance scores derived from the RR model were used to evaluate the relative contributions of seven digital technology tools in predicting teaching quality (Figure 3). The results showed that multimedia courseware had the highest predictive importance (0.0167 ± 0.0094), followed by e-textbooks (0.0142 ± 0.0080) and question banks (0.0129 ± 0.0087). In contrast, management software demonstrated a weaker in-

fluence (0.0091 ± 0.0129), while multimedia materials (0.0040 ± 0.0104) and websites (0.0013 ± 0.0125) made minimal contributions to the model's performance. Notably, subject software displayed a negative average contribution (−0.0005 ± 0.0087), suggesting it may function as a redundant or noisy feature in the prediction of teaching quality.

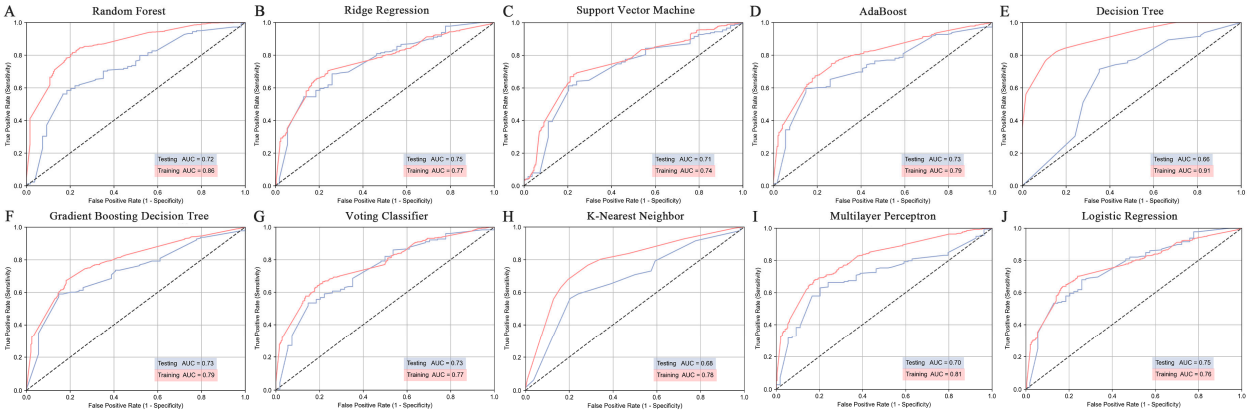

Figure 2. ROC curve analysis of ten machine learning models for predicting teaching quality. (**A**) Random Forest; (**B**) Ridge Regression; (**C**) Support Vector Machine; (**D**) AdaBoost; (**E**) Decision Tree; (**F**) Gradient Boosting Decision Tree; (**G**) Voting Classifier; (**H**) K-Nearest Neighbor; (**I**) Multilayer Perceptron; (**J**) logistic regression.

Table 1. Performance comparison of ten machine learning models across multiple classification metrics.

Characteristics	RF	RR	SVM	AB	DT	GBDT	VC	KNN	MLP	LR
AUC	0.72	0.75	0.71	0.73	0.66	0.73	0.73	0.68	0.70	0.75
Accuracy	0.78	0.78	0.77	0.77	0.71	0.76	0.75	0.70	0.74	0.76
Sensitivity/Recall	0.93	0.94	1.00	0.92	0.76	0.93	0.84	0.79	0.94	0.92
Specificity	0.26	0.24	0.00	0.28	0.52	0.22	0.46	0.41	0.07	0.26
FPR	0.74	0.76	1.00	0.72	0.48	0.78	0.54	0.59	0.93	0.74
FNR	0.07	0.06	0.00	0.08	0.24	0.07	0.16	0.21	0.06	0.08
PPV	0.81	0.80	0.77	0.81	0.84	0.80	0.84	0.82	0.77	0.80
NPV	0.54	0.54	0.00	0.50	0.40	0.48	0.46	0.37	0.27	0.48
F1 score	0.86	0.87	0.87	0.86	0.80	0.86	0.84	0.80	0.85	0.86

Note: RF, Random Forest; RR, Ridge Regression; SVM, Support Vector Machine; AB, AdaBoost; DT, Decision Tree; GBDT, Gradient Boosting Decision Tree; VC, Voting Classifier; KNN, K-Nearest Neighbor; MLP, Multilayer Perceptron; LR, logistic regression; AUC, area under the receiver operator curve; FPR, false positive rate; FNR, false negative rate; PPV, positive predictive value; NPV, negative predictive value.

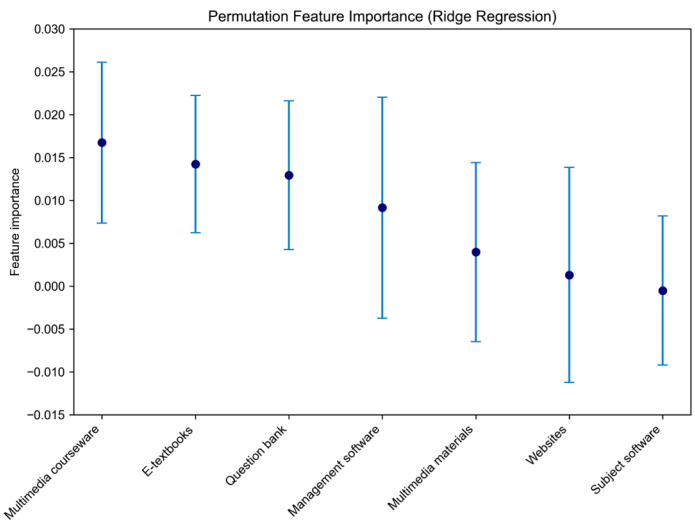

Figure 3. Permutation feature importance of digital technology tools in ridge regression model.

3.3.2. SHAP Analysis

To further clarify the contributions of various digital technology tools to teaching quality prediction in physical education, SHAP was employed for interpretability analysis. Figure 4A illustrates the average impact of each feature on predicted teaching quality across the full sample. Management software, websites, and multimedia materials showed strong positive effects, with higher values generally corresponding to increased predictions of teaching quality (as indicated by red rightward shifts). Figure 4B displays the cumulative contribution trajectories of individual features. The color gradient from blue to red indicates a progression from lower to higher predicted values. Figure 5A presents an SHAP force plot for the sample with the highest predicted teaching quality. In this case, aside from multimedia courseware and question banks (both used at frequency = 1), websites, subject software, and management software—when used more frequently—exerted stronger positive influences. Among these, websites emerged as the most impactful contributor. Figure 5B offers a detailed breakdown of feature-wise contributions, tracing the progression from the model's expected baseline prediction (E[f(x)] = 0.496) to the final predicted value of 1.162. The most substantial positive contributions came from websites (+0.26), multimedia courseware (+0.14), and question banks (+0.10), while multimedia materials had a slight negative effect (−0.01), and e-textbooks had negligible influence.

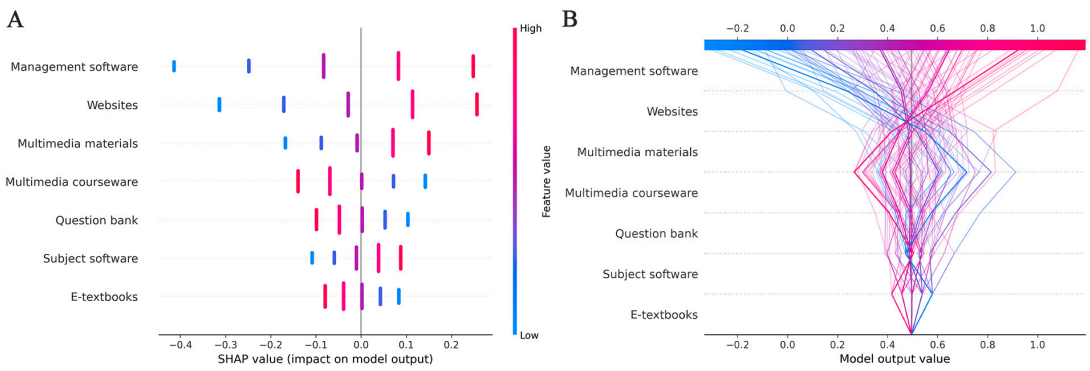

Figure 4. SHAP-based interpretation of feature contributions in the ridge regression model. (**A**) Summary plot; (**B**) SHAP decision plot.

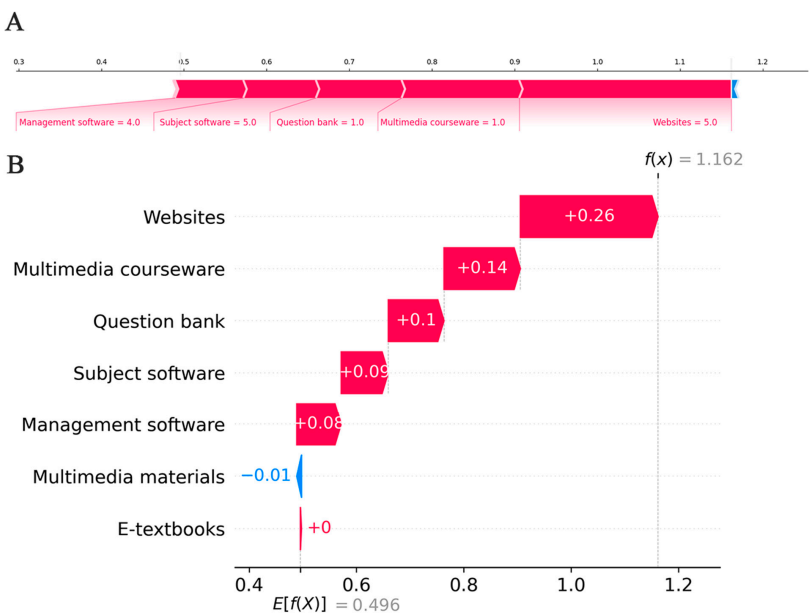

Figure 5. SHAP-based local interpretation of feature contributions for a high-scoring sample. (**A**) SHAP force plot; (**B**) SHAP waterfall plot.

3.3.3. The Impact and Synergistic Effects of Digital Technology Use on Teaching Quality

To further examine the relationship between digital technology use and teaching quality, visual analytic techniques such as predicted distribution plots and partial dependence plots (PDPs) were employed. As shown in Figures 6 and S4, predicted teaching quality was lowest when tool usage frequency fell within the [1, 2) range, increased within [2, 3), dipped slightly in [3, 4), and peaked at [4, 5]. This pattern suggests a possible diminishing return or inefficiency at moderate levels of use. Overall, higher usage frequency was positively associated with predicted teaching quality. PDP results revealed that management software, websites, and multimedia materials had the strongest positive effects. In contrast, multimedia courseware and question banks demonstrated diminishing marginal returns at higher usage frequencies, while subject software and e-textbooks showed limited influence.

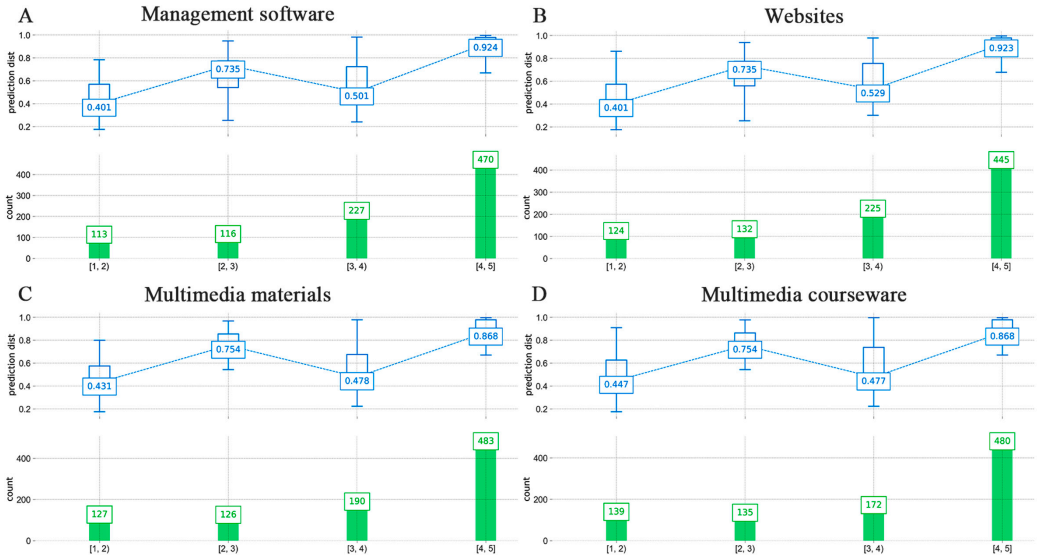

Figure 6. Distribution of predicted teaching quality scores across usage frequencies of four digital tools. (**A**) Management software; (**B**) Websites; (**C**) Multimedia materials; (**D**) Multimedia courseware. Boxplots show the predicted scores of teaching quality across different usage intervals, while bar charts indicate the number of corresponding samples in each interval.

Further analysis of synergistic effects (Figures 7 and S5) showed that management software, multimedia materials, subject software, and websites exhibited strong positive synergy when used in combination, contributing more to teaching quality than when used independently. Conversely, the frequent use of multimedia courseware, e-textbooks, and question banks was associated with a significant negative impact on predicted teaching quality.

3.4. Common Method Bias Test and Mediation Analysis

3.4.1. Common Method Bias Test

This study further investigated the underlying mechanisms through which digital technology use affects teaching quality. To assess potential common method bias, Harman's single-factor test was applied to all 69 items from four separate questionnaires. The results showed that seven factors had eigenvalues greater than 1, and the variance explained by the first factor after rotation was 27.47%, which is below the critical threshold of 40%, suggesting that common method bias was not a significant issue in this study. The Pearson correlation analysis revealed significant positive correlations among digital technology use, perceived teacher support, academic self-efficacy, and teaching quality (all $p < 0.01$; see Table 2).

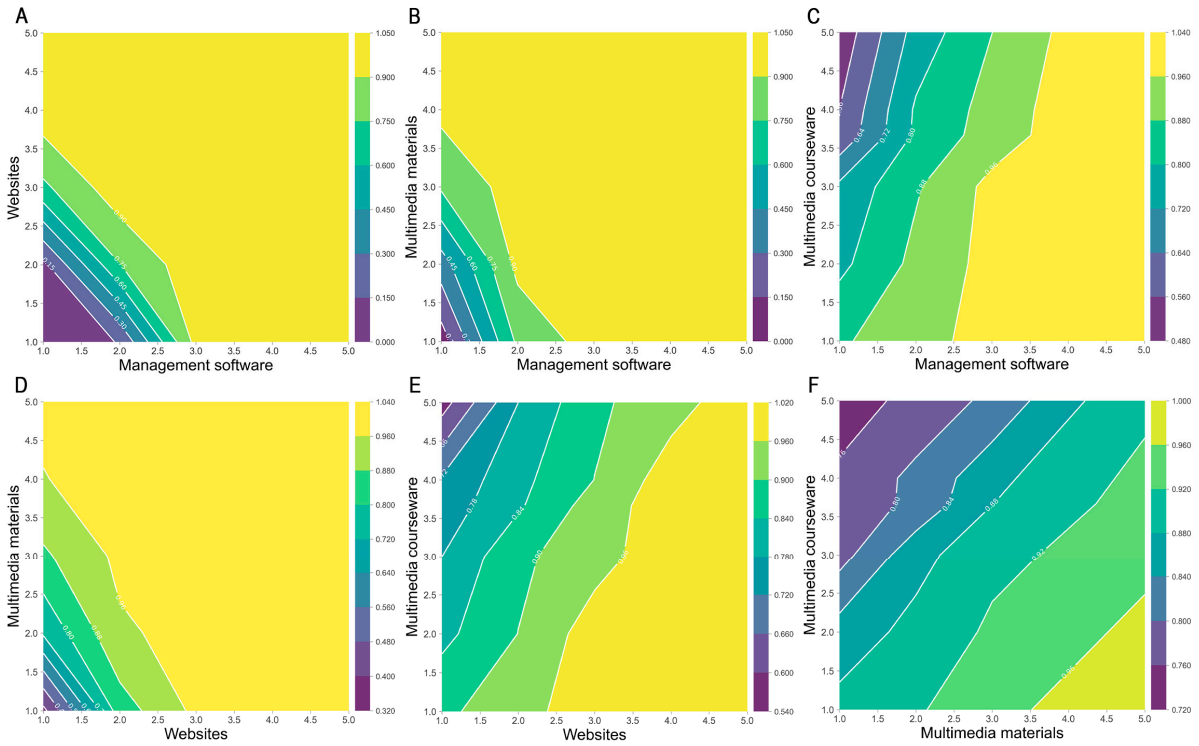

Figure 7. Interaction effects of paired digital tools on predicted teaching quality scores. (**A**) Management software and Websites; (**B**) Management software and Multimedia materials; (**C**) Management software and Multimedia courseware; (**D**) Websites and Multimedia materials; (**E**) Websites and Multimedia courseware; (**F**) Multimedia materials and Multimedia courseware.

Table 2. Correlation analysis.

		Mean	SD	1	2	3	4
1	Digital technology use	29.827	10.707	1			
2	Perceived teacher support	69.006	10.390	0.463 **	1		
3	Academic self-efficacy	83.867	12.400	0.478 **	0.695 **	1	
4	Teaching quality	99.387	17.558	0.497 **	0.647 **	0.590 **	1

Note: ** $p < 0.01$.

3.4.2. Mediation Effect Analysis

To explore the mediating roles of perceived teacher support and academic self-efficacy, Models 4 and 6 of the PROCESS macro in SPSS were used, applying a bias-corrected percentile bootstrap method with 10,000 resamples. The results are shown in Table 3. Based on the significant direct effect of digital technology use on teaching quality, three indirect pathways were tested: Digital technology use → perceived teacher support → teaching quality; Digital technology use → academic self-efficacy → teaching quality; Digital technology use → perceived teacher support → academic self-efficacy → teaching quality. The estimated effect sizes for these mediation pathways were 0.303, 0.068, and 0.096, respectively. The corresponding 95% confidence intervals were [0.238, 0.374], [0.038, 0.103], and [0.061, 0.134], all of which excluded zero, indicating statistical significance. These findings demonstrate that perceived teacher support and academic self-efficacy function not only as independent mediators but also as function as a chain mediating effect. Therefore, Hypotheses 2, 3, and 4 were supported.

Table 3. Mediation effect analysis.

	Path	Effect Value	Boot SE	Boot LLCI	Boot ULCI	*p*
Direct effect	Digital technology use → Teaching quality	0.347	0.040	0.268	0.426	0.000
Indirect effect	Digital technology use → Perceived teacher support → Teaching quality	0.303	0.035	0.238	0.374	0.000
	Digital technology use → Academic self-efficacy → Teaching quality	0.068	0.017	0.038	0.103	0.000
	Digital technology use → Perceived teacher support → Academic self-efficacy → Teaching quality	0.096	0.019	0.061	0.134	0.000
Total effect	Digital technology use → Teaching quality	0.814	0.043	0.731	0.898	0.000

4. Discussion

This study constructed interpretable prediction models based on machine learning techniques to investigate, for the first time, the contributions of multiple digital technology tools to teaching quality in university physical education classes. Simultaneously, SEM was employed to elucidate the underlying mechanisms by which digital technology use, perceived teacher support, and academic self-efficacy influence teaching quality in this context. The findings demonstrate that the RR model delivered superior predictive performance, identifying multimedia courseware, e-textbooks, and question banks as the most influential features in the teaching quality prediction model. The SHAP analysis revealed that management software, websites, and multimedia materials were the core positive predictors, showing significant positive associations with teaching quality. Interestingly, the frequent use of multimedia courseware, question banks, and e-textbooks was found to have a negative impact on teaching quality in physical education settings. This suggests that tools traditionally effective in academic contexts may not translate directly to subjects that emphasize physical interaction and experiential learning. Moreover, an analysis of synergistic effects indicated that combinations of management software, multimedia materials, subject software, and websites yielded significant positive joint effects on teaching quality when used in tandem, underscoring the importance of a strategic integration of complementary digital tools. In addition, the study demonstrated that digital technology use influences teaching quality indirectly through two key mediating pathways: perceived teacher support and academic self-efficacy. These findings validate the chain mediation model, emphasizing that the effectiveness of digital technology in education is not only a function of tool selection, but also of how such tools enhance teacher–student interactions and foster students' confidence in their learning capabilities. By integrating interpretable machine learning models with structural equation modeling, this research provides a robust framework for identifying and understanding the key drivers of teaching quality in digitalized physical education classrooms. The results offer both theoretical insights and practical implications for optimizing the deployment of digital technologies and formulating evidence-based strategies for enhancing instructional effectiveness in higher education.

This study is the first to employ interpretable machine learning methods to identify that the use of management software, websites, and multimedia materials in physical education classrooms positively influences teaching quality. Moreover, the combined use of management software, multimedia materials, subject software, and websites produced a stronger synergistic effect in enhancing teaching outcomes. These findings are consistent with previous research showing that integrating diverse digital tools and online resources into classroom instruction fosters more personalized learning experiences and collabora-

tive environments, thereby increasing student engagement and participation [42]. In the context of physical education, digital technologies have been shown to improve student involvement and physical fitness outcomes [43]. Specifically, tools such as management software, websites, and multimedia materials support real-time assessment and feedback, enhance teacher–student interaction, and improve learning efficiency and instructional effectiveness [44]. Online resources like websites and multimedia content also enhance the accessibility, equity, and democratization of educational materials, reducing reliance on traditional textbooks [45]. Furthermore, using video- and audio-based multimedia to deliver instructional content improves students' comprehension and retention—particularly beneficial in subjects such as physical education, which emphasize visual and experiential learning [46]. However, a key limitation of the current analytical approach is its restriction to evaluating the synergistic effects of only two variables at a time on teaching quality, without capturing the cumulative impact of multiple interacting factors. Future studies should consider employing more advanced methodological frameworks to overcome this limitation.

Notably, this study found that the frequent use of multimedia courseware, question banks, and e-textbooks in physical education classes was associated with a negative impact on teaching quality. This finding contrasts with previous research, which has consistently reported that multimedia courseware, question banks, and e-textbooks enhance learning outcomes in subjects such as science, language, and mathematics. In contrast, the present study identifies an opposing trend within the domain of physical education. While these tools effectively support cognitive comprehension and assessment readiness in knowledge-based disciplines, their applicability may not extend to performance-oriented fields like physical education, which demand active bodily engagement and context-dependent learning. Physical education, in particular, emphasizes experiential practice and the development of motor skills through direct physical involvement. Overreliance on multimedia courseware and e-textbooks for delivering theoretical content or demonstrating skills may reduce opportunities for hands-on training, thereby limiting physical participation and undermining students' interest and experiential satisfaction [47]. While question banks are effective for reinforcing knowledge and improving test performance in academic subjects, their use in physical education may overemphasize written assessment at the expense of practical skill development. This shift can undermine students' intrinsic motivation and, ultimately, degrade the overall quality of instruction [48]. Accordingly, it is recommended that the use of multimedia courseware and e-textbooks for delivering theoretical content in physical education be carefully regulated in both frequency and duration to prevent cognitive overload and student disengagement. Instead, multimedia tools should be purposefully employed to visualize technical skills and complex movements, while ensuring sufficient time is preserved for hands-on, experiential learning. Such an approach can enhance student interest and improve overall instructional quality. By identifying specific digital technologies that positively or negatively influence teaching quality in physical education, this study provides a more nuanced understanding of the interaction between subject content and technological tools. In doing so, it addresses a gap in the literature, which has largely focused on academic subjects or general multimedia use without accounting for subject-specific pedagogical contexts.

This study further revealed that the use of digital technology in physical education classrooms not only has a direct positive effect on teaching quality but also enhances it indirectly by promoting perceived teacher support and academic self-efficacy. This finding is consistent with prior research [49]. When teachers effectively and consistently integrate digital tools—such as classroom management software and educational websites—into instruction to enhance guidance, interaction, and emotional engagement, students are more

likely to perceive strong teacher support. In turn, this perception fosters greater learning motivation and classroom participation [50], both of which are key determinants of teaching quality. Thus, the effective use of digital technologies strengthens students' perceptions of teacher support, thereby contributing to improved instructional outcomes. Moreover, in line with previous research, this study confirmed a significant positive association between perceived teacher support and academic self-efficacy, and further demonstrated that these two factors jointly serve as a sequential mediating mechanism between digital technology use and teaching quality. Teacher-provided emotional, academic, and competence-related support has been strongly linked to students' active engagement and increased academic confidence [51]. Students with higher levels of academic self-efficacy tend to show greater persistence and initiative in their studies, resulting in higher engagement and improved instructional outcomes [52]. These findings align with the principles of social cognitive theory, which posits that perceived teacher support enhances academic self-efficacy, ultimately leading to improved learning performance and teaching quality [28]. Therefore, as digital transformation continues in physical education, educators should not only consider when and how frequently to use various digital tools, but also prioritize leveraging them to strengthen academic support and teacher–student interaction. This approach can enhance students' self-efficacy and help ensure that the integration of digital technologies leads to meaningful improvements in teaching quality.

This study is the first to integrate interpretable machine learning methods with structural equation modeling to examine both the predictive factors and underlying mechanisms influencing teaching quality in physical education classrooms. While most prior research on the educational use of digital technology has concentrated on academic subjects such as mathematics and reading, the present findings provide compelling evidence that the effective integration of digital tools can also significantly enhance teaching quality in practice-intensive subjects like physical education. Notably, the study underscores the distinct pedagogical nature of physical education, revealing that certain digital tools—previously shown to be effective in academic contexts—may negatively affect teaching quality when applied in this setting. These findings suggest that, in practical implementation, educational administrators and instructors should prioritize high-impact digital tools—such as management software and multimedia materials—that align more closely with the experiential and interactive demands of physical education. These technologies can enrich instructional content and foster more dynamic classroom interactions. Additionally, the study highlights the critical mediating roles of perceived teacher support and academic self-efficacy in digitally enhanced learning environments. This calls for a dual emphasis in educational practice: alongside technological adoption, educators must actively sustain academic guidance and emotional support to ensure students feel engaged and receive personalized feedback. Such support is vital for maintaining students' academic self-efficacy in digital learning settings. From a policy perspective, educational leaders should aim to strike a balance between investments in digital "hardware" and "software". Beyond infrastructure and platform development, equal attention must be given to the creation of high-quality digital instructional content and the enhancement of teachers' digital pedagogical competencies. Only through this comprehensive strategy can the full potential of digital technologies be translated into meaningful improvements in teaching quality.

Nevertheless, several limitations of this study should be acknowledged. First, the cross-sectional design constrains causal inference and limits the ability to capture temporal dynamics. Future research should consider employing longitudinal designs to track developmental trajectories, or experimental approaches—such as randomized controlled trials—to examine causal relationships among digital technology use, academic self-efficacy, and teaching quality under controlled conditions. Second, the study relies entirely on self-

reported questionnaire data, which may introduce common method bias and subjectivity. Although validated scales, assurances of anonymity, and robustness checks were employed to mitigate these issues, social desirability bias and other distortions cannot be fully eliminated. Future studies could adopt multi-method approaches, incorporating objective indicators such as academic performance scores, physiological data (e.g., metrics from wearable devices), and structured classroom observation protocols. Moreover, combining student self-reports with teacher evaluations or peer assessments may help reduce single-source bias. Third, this study utilized convenience and snowball sampling, with all participants recruited from Chinese universities. Importantly, over half of the sample comprised first-year undergraduates, potentially introducing experience-related bias and limiting the overall representativeness. To improve the generalizability and applicability of future research, we recommend adopting probability-based random sampling and incorporating cross-cultural comparative designs. Moreover, future studies should aim for more balanced sampling across academic year levels or implement stratified analyses based on year group. Fourth, the valid response rate of 71.84% suggests the potential for survivorship or nonresponse bias. Subsequent research should incorporate strategies for tracking and comparing respondents with non-respondents. Fifth, the analytical models—both machine learning and structural equation modeling—did not account for the differential weighting of variable subdimensions, potentially limiting item-level resolution. Future research should employ more refined measurement modeling techniques to address this limitation. Sixth, although age, gender, academic year, ethnicity, and location were included as control variables, their possible moderating effects were not examined. Future studies should consider multigroup analysis or interaction modeling to explore how these variables might influence the observed associations. Finally, this study focused on only seven types of digital technologies, academic self-efficacy, and teaching quality, using validated yet relatively established measurement instruments. Given the rapid evolution of educational technologies and pedagogical models, future research should consider developing or adopting up-to-date measurement tools that more accurately reflect contemporary digital learning environments. For example, integrating instruments tailored to hybrid instruction, gamification, AI-driven platforms, or wearable technologies could provide a more nuanced understanding of digital engagement and learning outcomes.

5. Conclusions

This study is the first to integrate interpretable machine learning methods with structural equation modeling to systematically examine the contributions and underlying mechanisms of digital technology use on teaching quality in university-level physical education. The findings indicate that management software, websites, and multimedia materials serve as key positive predictors of teaching quality in this context. In contrast, the frequent use of multimedia courseware, question banks, and e-textbooks was associated with negative effects on teaching quality. Additionally, significant positive synergistic effects emerged when management software, multimedia materials, subject software, and websites were used in combination, underscoring the importance of strategically integrating complementary digital tools. The study further demonstrated that digital technology use influences teaching quality not only directly but also indirectly through the mediating roles of perceived teacher support and academic self-efficacy. By shifting the research focus beyond traditional academic disciplines, this study addresses a critical gap in the literature and offers new empirical evidence on the function of digital tools within the distinct pedagogical context of physical education. In doing so, it provides both theoretical insights and practical recommendations for optimizing digital technology integration in physical

education, thereby contributing to the advancement of evidence-based digital pedagogy in higher education.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/app15147689/s1. Figure S1: Lasso regression analysis. (A) Coefficient path plot, (B) Cross-validation error curve for regularization parameter (λ). Figure S2: Learning curve of the random forest classifier for teaching quality prediction. Figure S3: Comparison of ridge regression-based confusion matrices for quality of teaching across digital technology features. Figure S4: Predicted teaching quality distribution across usage frequencies of digital technology tools. (A) Question bank, (B) Subject software, and (C) E-textbooks. Figure S5: Contour maps of synergistic effects on predicted teaching quality across pairs of digital technology tools. Table S1: Details of hyperparameter for ML models. Table S2. Assessment of multicollinearity among model variables using variance inflation factors (VIF).

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Abbreviations

The following abbreviations are used in this manuscript:

χ2/df	Chi-square to Degrees of Freedom Ratio
GFI	Goodness-of-Fit Index
AGFI	Adjusted Goodness of Fit Index
CFI	Comparative Fit Index
TLI	Tucker–Lewis Index
RMSEA	Root Mean Square Error of Approximation
NFI	Normed Fit Index
IFI	Incremental Fit Index
RFI	Relative Fit Index
CI	Confidence interval
RF	Random forest
RR	Ridge regression
SVM	Support vector machine
AB	AdaBoost

DT	Decision tree
GBDT	Gradient boosting decision tree
VC	Vting classifier
KNN	K-nearest neighbors
MLP	Multilayer perceptron
LR	Logistic regression
LASSO	Least absolute shrinkage and selection operator
VIF	Variance inflation factor
SEM	Structural equation model
ANOVA	Analysis of variance
ROC	Receiver operating characteristic
PDPs	Partial dependence plots

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